List of publications

Satvik Singh

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Selected key publications

Three selected research papers spanning entanglement theory, information transfer and the limits of noisy quantum memories.

1. S. Singh and N. Datta. **Information storage and transmission under Markovian noise.** *PRX Quantum* 7, 020312 (2026).
   https://doi.org/10.1103/8rnj-667b

   Main achievements. Determines long-time classical and quantum transmission capacities of finite-dimensional Markovian noise from its peripheral structure, proving strong additivity and strong converses. Provides tight, efficiently computable storage bounds and quantitative convergence times. For typical independent qubit noise with fixed, time-independent error-correction, establishes memory-lifetime upper bounds scaling at most exponentially with the number of qubits under global correction and logarithmically under local correction.

2. S. Singh. **Gaussian mean width strong converse bound on the classical identification capacity of quantum channels.** Preprint, arXiv:2606.05032 (2026).
   https://arxiv.org/abs/2606.05032

   Main achievements. Develops a geometric framework for establishing the tightest strong converse bounds to date on classical identification via quantum channels, using tools from high-dimensional probability and geometric analysis. Building on this framework, a subsequent paper (item 17) exactly characterizes the identification capacity of sufficiently low-noise channels and yields the first example of strict superadditivity of this capacity. These results establish fundamental limits on communication when receivers only need to test the identity of a message rather than reconstruct it in full.

3. S. Singh and I. Nechita. **The PPT² conjecture holds for all Choi-type maps.** *Annales Henri Poincaré* 23, 3311-3329 (2022).
   https://doi.org/10.1007/s00023-022-01166-0

   Main achievements. Proves that composing two positive-partial-transpose (PPT) maps with diagonal unitary covariance yields an entanglement-breaking map, establishing the PPT² conjecture for this broad class in arbitrary dimension. Develops factor-width methods for pairwise completely positive matrices. The result constrains entanglement distribution through quantum repeaters using symmetric PPT resources.

Other published articles

4. A. Gulati, I. Nechita and S. Singh. Entanglement in cyclic sign invariant quantum states. *Journal of Mathematical Physics* 66, 122202 (2025).
   https://doi.org/10.1063/5.0285294

5. S. Singh, M. Rahaman and N. Datta. Zero-error communication under discrete-time Markovian dynamics. *Quantum* 9, 1910 (2025).
   https://doi.org/10.22331/q-2025-11-14-1910

6. S. Singh and S. Strelchuk. Simultaneous superadditivity of the direct and complementary channel capacities. *IEEE Transactions on Information Theory* 71(11), 8542-8550 (2025).
   https://doi.org/10.1109/TIT.2025.3602136

7. S. Singh, N. Datta and I. Nechita. Ergodic theory of diagonal orthogonal covariant quantum channels. *Letters in Mathematical Physics* 114, 121 (2024).
   https://doi.org/10.1007/s11005-024-01864-2

8. S. Singh and N. Datta. Fully undistillable quantum states are separable. *IEEE Transactions on Information Theory* 69(10), 6564-6570 (2023).
   https://doi.org/10.1109/TIT.2023.3285444

9. S. Singh. A brief discourse on Hausdorff dimension and self-similarity. *The American Mathematical Monthly* 129(9), 831-845 (2022).
https://doi.org/10.1080/00029890.2022.2105117

10. S. Singh and N. Datta. Coherent information of a quantum channel or its complement is generically positive. *Quantum* 6, 775 (2022).
https://doi.org/10.22331/q-2022-08-11-775

11. S. Singh and I. Nechita. Diagonal unitary and orthogonal symmetries in quantum theory: II. Evolution operators. *Journal of Physics A: Mathematical and Theoretical* 55, 255302 (2022).
https://doi.org/10.1088/1751-8121/ac7017

12. S. Singh and N. Datta. Detecting positive quantum capacities of quantum channels. *npj Quantum Information* 8, 50 (2022).
https://doi.org/10.1038/s41534-022-00550-2

13. S. Singh and I. Nechita. Diagonal unitary and orthogonal symmetries in quantum theory. *Quantum* 5, 519 (2021).
https://doi.org/10.22331/q-2021-08-09-519

14. S. Singh. Entanglement detection in triangle-free quantum states. *Physical Review A* 103, 032436 (2021).
https://doi.org/10.1103/PhysRevA.103.032436

15. I. Nechita and S. Singh. A graphical calculus for integration over random diagonal unitary matrices. *Linear Algebra and its Applications* 613, 46-86 (2021).
https://doi.org/10.1016/j.laa.2020.12.014

Other preprints

16. R. Salzmann and S. Singh. Transmitting algebras through quantum channels. Preprint (2026).
https://arxiv.org/abs/2609.40054

17. S. Singh. The entanglement-assisted transmission capacity is a strong converse bound for identification. Preprint (2026).
https://arxiv.org/abs/2608.11000

18. S. Singh. A short proof of the modified Kretschmann-Schlingemann-Werner conjecture. Preprint (2026).
https://arxiv.org/abs/2606.16418

19. S. Singh and B. Bergh. Discriminating idempotent quantum channels. Preprint (2026).
https://arxiv.org/abs/2603.28582

20. A. Müller-Hermes and S. Singh. Bi-PPT channels are entanglement breaking. Preprint (2022; revised 2023).
https://arxiv.org/abs/2204.01685

PhD thesis

21. S. Singh. Detection and estimation of quantum channel capacities. PhD thesis, University of Cambridge (2025).
https://doi.org/10.17863/CAM.122710